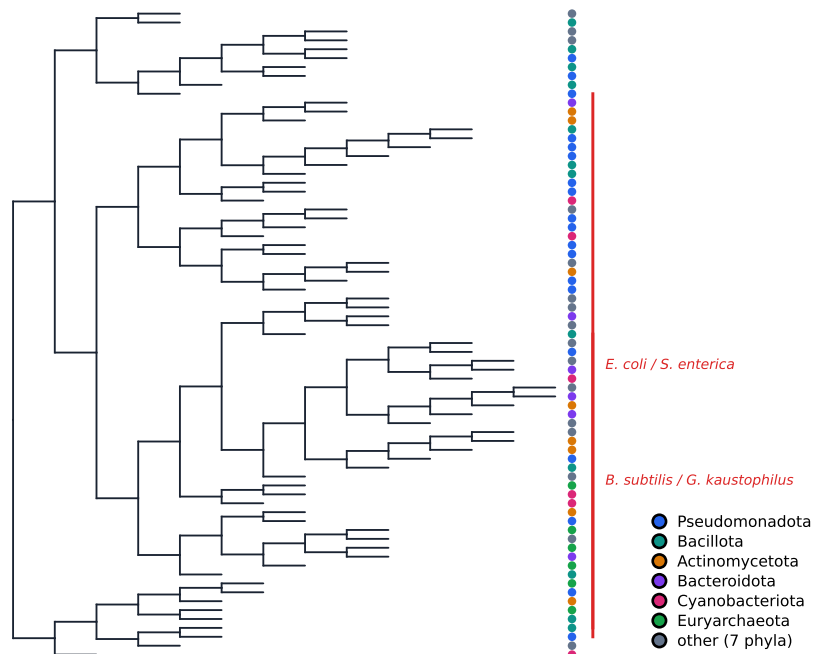


Figure 6. Real-data validation

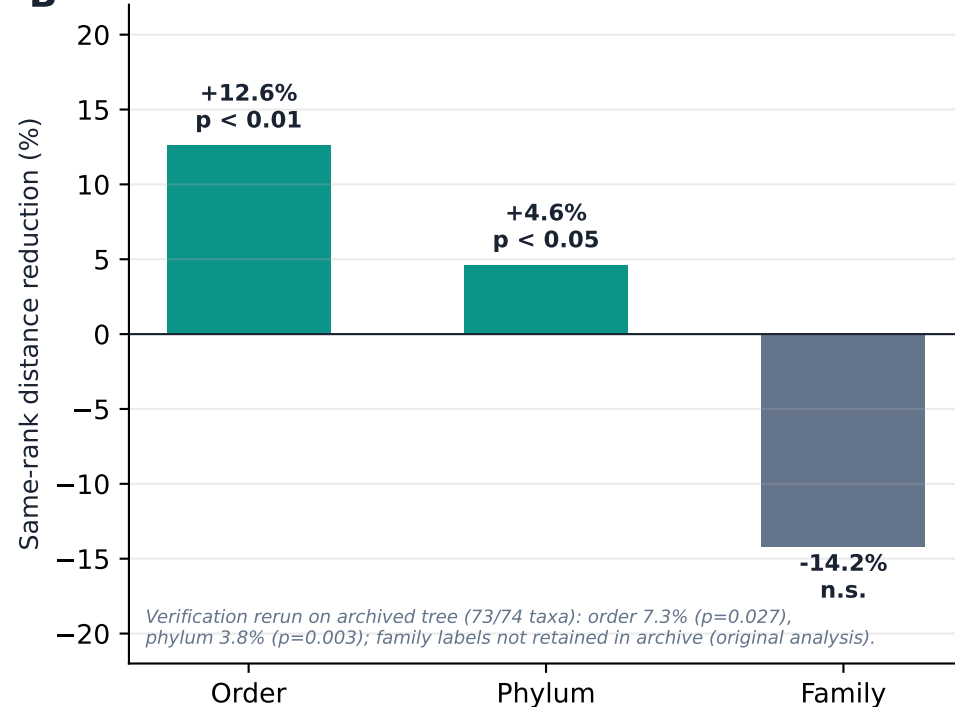
A

16S rRNA tree, 73 taxa (k=5,gap2; 1.2 s, no alignment)



B

Phylogenetic signal by taxonomic rank



C

Fusang v1 (2023) → Tardigrade Edition

v1: deep learning

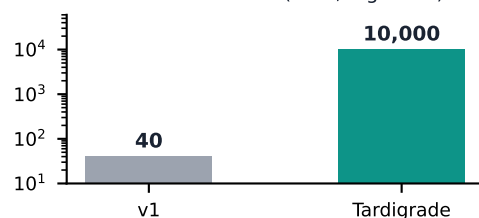


re-architecture:
no GPU, no training, no Docker

Tardigrade: k-mer frequency vectors



max dataset size (taxa, log scale)



D

Gene-length DNA benchmark (13 mtDNA loci, 25 fish)

